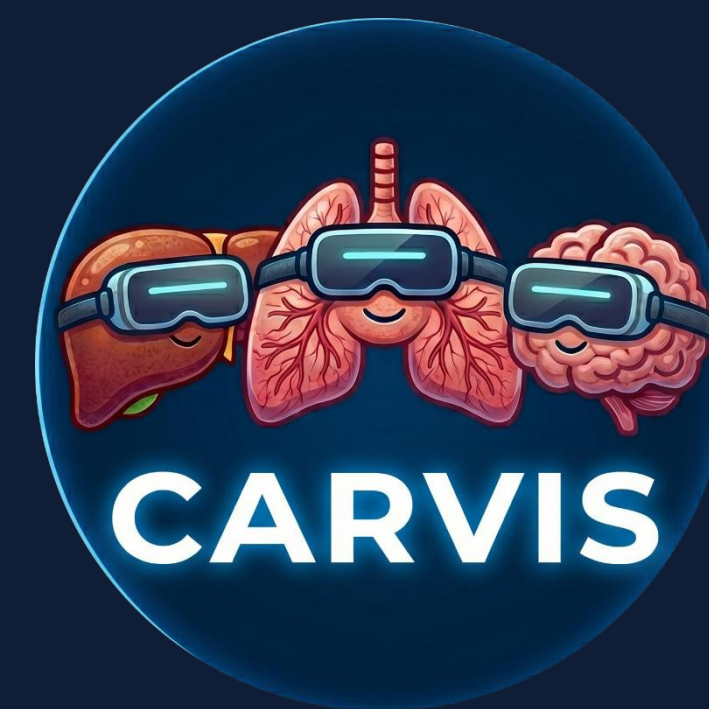




CARVIS

Clinical Anatomical Rendering & Visualization Intelligent System

Alperen Berke Çetinkaya · Muhammed Yusuf Özcan · Sezer Atas · Mete Serpil
Advisor: Gül Tokdemir | Çankaya University · Dept. of Computer Engineering



The Problem

Medical imaging workflows are often fragmented: segmentation, measurement, reporting, and case review are handled in separate tools, creating context-switching, weak traceability, and privacy concerns with cloud-based services. Clinicians also rely on 2D CT and MRI slices, which are hard to interpret and limit spatial understanding. CARVIS addresses these gaps with an integrated, offline workflow that produces 3D segmentations and enables volume rendering of MRI and CT data on Meta Quest 3, providing measurable, explainable, and patient-specific clinical insights in an immersive 3D environment.

Our Solution

CARVIS is an offline platform that enables the generation and examination of patient-specific 3D digital twins of organs through VR goggles, with AI-supported reporting. From NIFTI/DICOM inputs, it performs nnU-Net-based 3D segmentation, lesion-centric volumetrics, and atlas-backed anatomical localization. Findings are delivered through a case-specific, evidence-linked Clinical Q&A system and automatically generated structured reports. All outputs are visualized in 3D on Meta Quest 3 via volume rendering, enabling an immersive case review. All processing runs locally, with no cloud transfer.

System Workflow

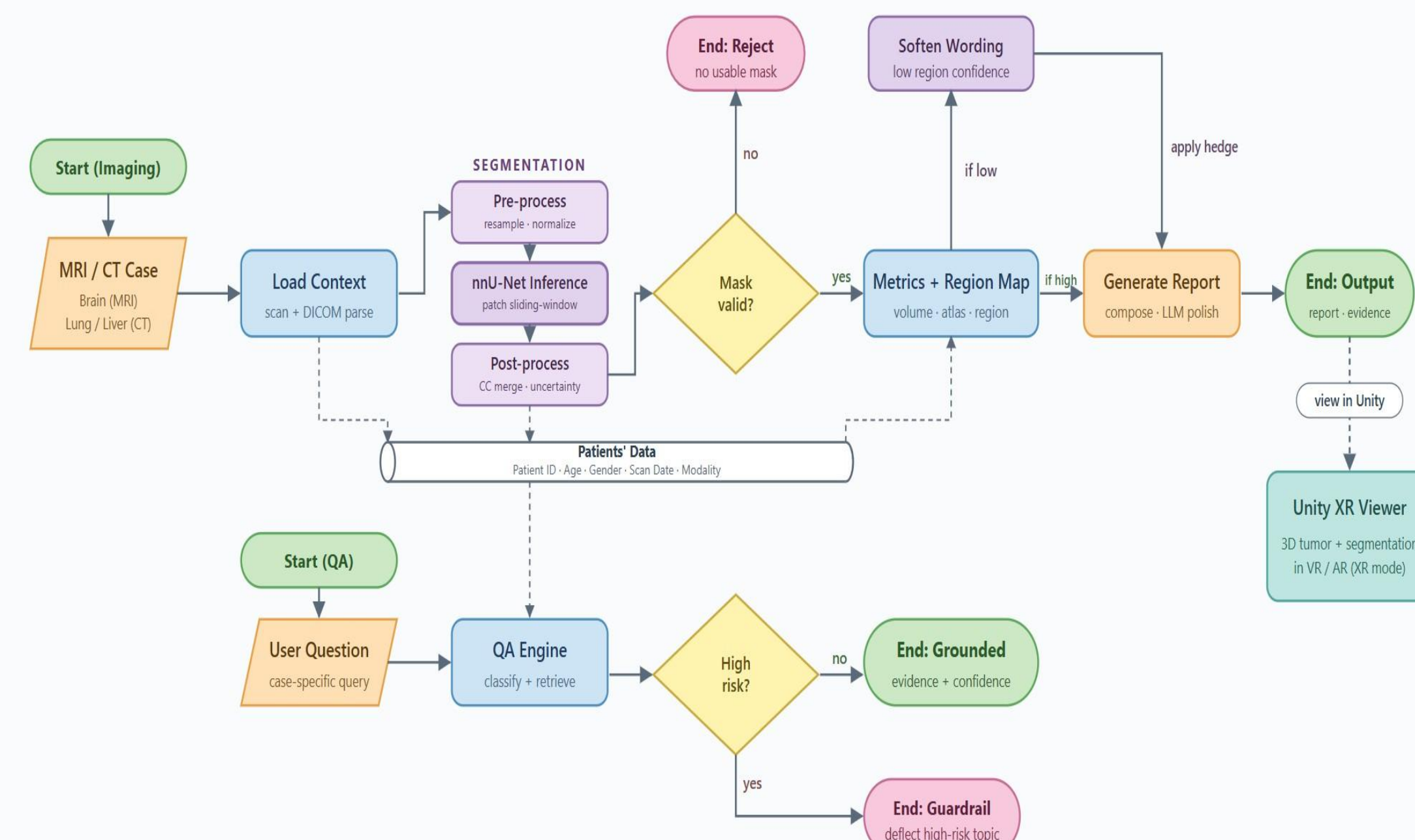


Figure 5. System Workflow.

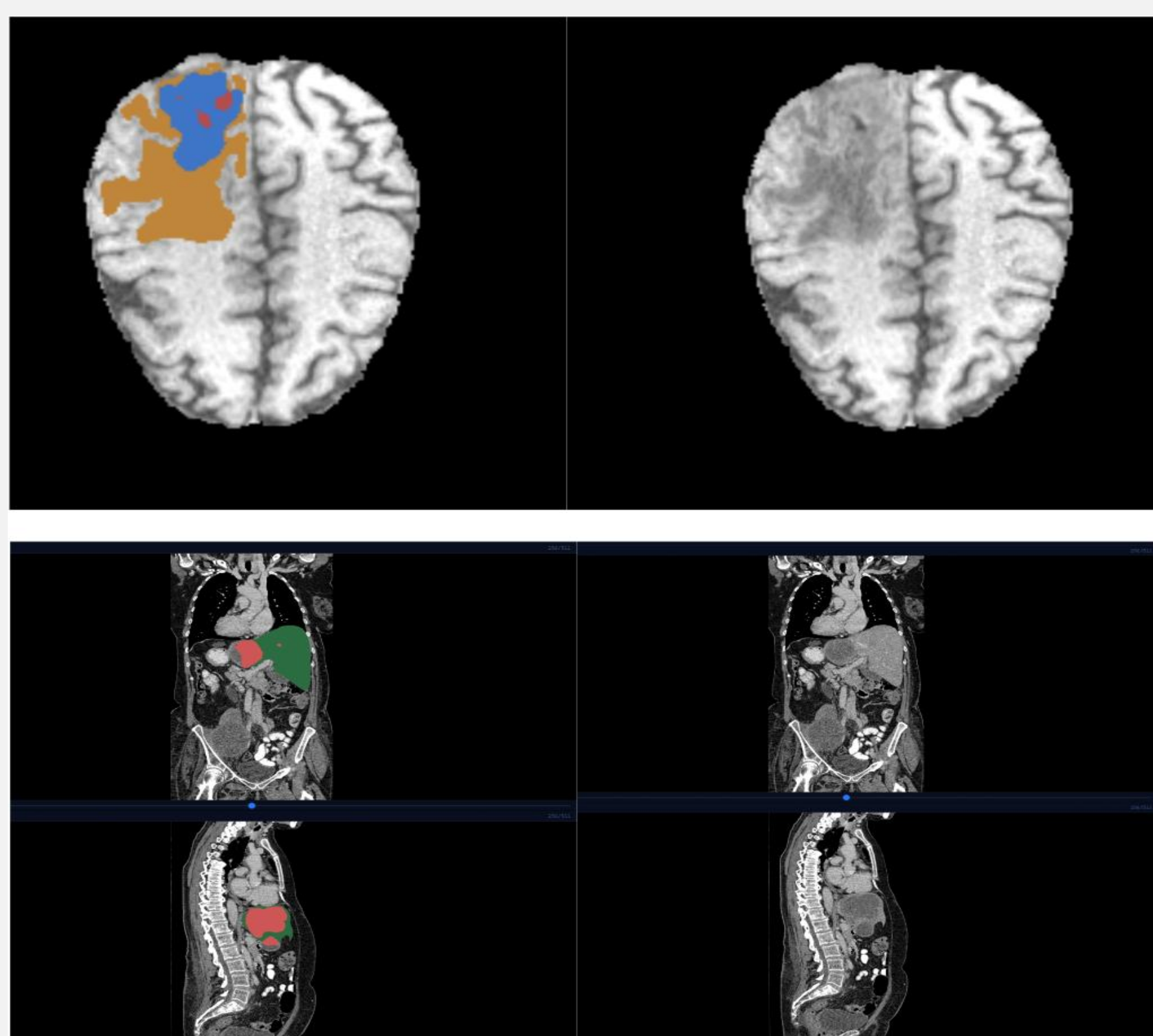


Figure 1. Brain MRI and liver CT viewers with tumor overlays.

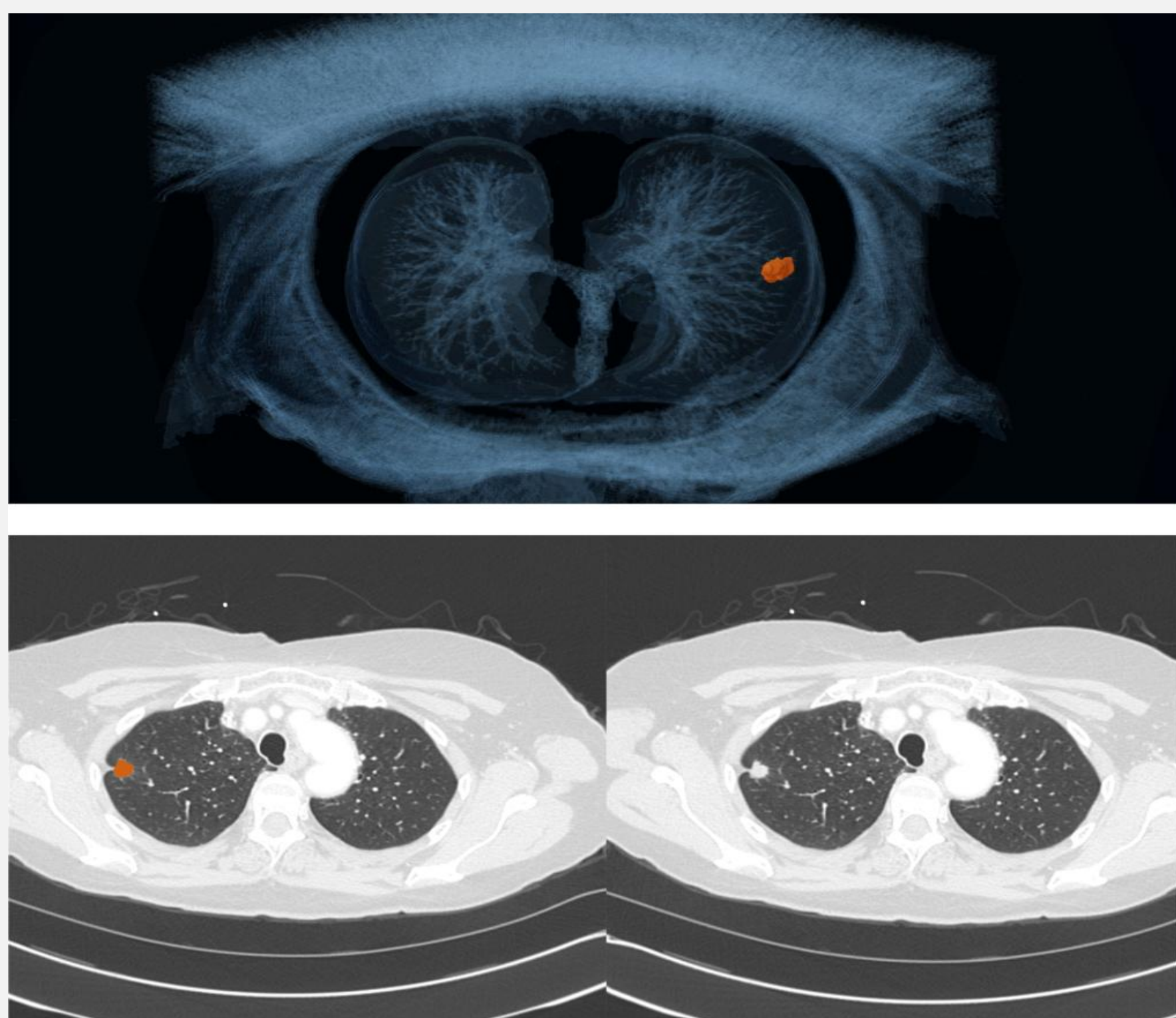


Figure 3. 3D mesh and CT viewers of lung tumor.

Technology Used

Desktop App: React 18, Electron, FastAPI, Pydantic, Python

Segmentation: nnU-Net v2-v1, SimpleITK, PyDICOM, Nibabel, NumPy, Nilearn, Harvard-Oxford Atlas

Q&A & Report Generation: Qwen 2.5 (GGUF) via llama.cpp, RapidFuzz, PySpellChecker, OpenAI-compatible local inference endpoint

Unity / VR: Meta XR Tools, adb, Unity Volume Rendering (mlavik1)

Results

CARVIS was evaluated on BraTS 2020 brain MRI cases, with preliminary liver and lung CT tests. It consistently generated patient-specific lesion volumetrics, lesion counts, maximum diameter, atlas-backed localization, structured report drafts, and evidence-linked Clinical Q&A outputs, all successfully visualized in 3D on Meta Quest 3 via volume rendering. Internal evaluation showed complete report sections, traceable answers, and no unsupported clinical claims under the project's groundedness checks. These results demonstrate a coherent offline pipeline for segmentation-based tumor case analysis, without claiming clinical diagnostic accuracy.



Figure 2. CARVIS XR demo.

Conclusion

CARVIS demonstrates a coherent offline platform that turns MRI and CT segmentation outputs into measurable, anatomically localized, and explainable patient-specific clinical artifacts. By combining 3D segmentation, quantitative metrics, structured reporting, evidence-linked Clinical Q&A, and immersive 3D visualization on Meta Quest 3 in a single local runtime, the platform provides a strong foundation for future clinical validation, expanded organ coverage, PACS integration, and multimodal imaging support.

Clinical Context	
Study date is unavailable. Clinical note is not available in the current package. Prior comparison material is not available.	
• Patient ID:	BraTS20_Validation_001
• Study Date:	Unknown
• Modality:	MR (FLAIR, T1, T1ce, T2 sequences)
• Available Imaging Inputs:	0 DICOM series directories and 4 NIFTI files.
• Clinical note present:	no; prior reports present: no.
• Report detail mode:	balanced.
• Purpose:	case-level brain tumor imaging assessment with automated segmentation review.
Findings	
The dominant pattern localizes to Left Occipital lobe, subcortical, middle level. Overall, the segmentation supports multiple lesion groups with fragmented internal segmentation components. Approximate regional overlap is concentrated in Left Frontal Pole, Left Cerebral White Matter, Left Temporal Pole. The qualitative burden pattern is predominantly non-enhancing/infiltrative with limited edema burden, while any mass effect or midline shift statements should be treated as secondary automated flags rather than direct measurements.	
• Total segmented tumor burden:	117.93 cm3 (117932.00 mm3). Confidence: Moderate (measurement uncertainty: +/-3.0 %). 95 % CI for total volume: 114.39-121.47 cm3. Maximum segmented diameter: 83.00 mm.
• Lesion architecture after class merge:	3 lesion groups across 8 segmented components.
• Component composition:	enhancing 19.95 cm3 (16.9 %), non-enhancing 97.78 cm3 (82.9 %), edema 0.32 cm3 (0.3 %).
• Dominant segmented component:	non-enhancing burden at 97.78 cm3 (82.9 % of total).
• Imaging phenotype:	predominantly non-enhancing/infiltrative with limited edema burden.
• Lesion #7 (dominant component):	nonenhancing component measuring 91933.00 mm3 (78.0 % of total burden) with max diameter 83.00 mm, centered in Left Occipital lobe, subcortical, middle level.
LOCALIZATION	
• Anatomic location:	Left Occipital lobe, subcortical, middle level.

Figure 4. Clinical report with evidence-based references.

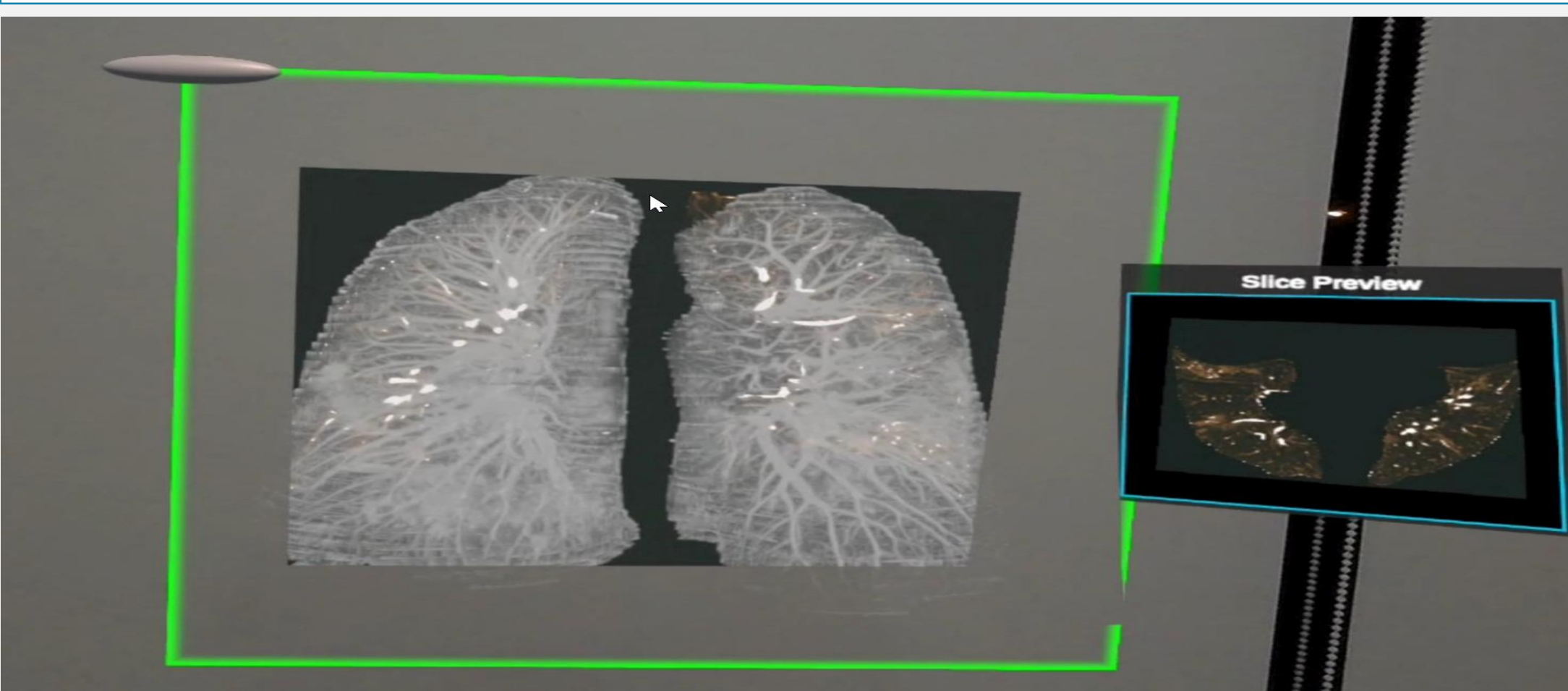


Figure 6. Meta Quest 3 XR view with slicing plane.

Contributors

We thank our advisor Gül Tokdemir for her guidance and the Çankaya University Dept. of Computer Engineering for academic support.



Figure 7. Advisor & Team.



Scan for full project resources.

